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EXPANDABLE INTRODUCER

Abstract

An expandable introducer includes an expandable introducer sheath coupled to an extending distally from an introducer hub via a connector. A distal portion of the introducer sheath includes a tip coupled to the sheath, the tip including a first circumferential portion and a second circumferential portion, the second circumferential portion being more flexible than the first circumferential portion. The second circumferential portion of the tip is circumferentially aligned with a folded portion of the sheath.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present application claims the benefit of U.S. Provisional Application No. 63/560,074, filed Mar. 1, 2024, and U.S. Provisional Application No. 63/675,889, filed Jul. 26, 2024, the contents of each of which are incorporated by reference herein in their entirety.

FIELD OF THE INVENTION

[0002] The present invention relates to an expandable introducer device for providing percutaneous access to a patient's vasculature for a transcatheter device delivery system.

BACKGROUND

[0003] An introducer device is used to provide percutaneous access to the vascular system of a patient and functions to permit the introduction and positioning of various minimally invasive delivery devices within the patient's vasculature. In general, making the smallest incision is the most desirable and leads to less complications, less trauma, and improved patient outcomes. Some delivery devices, however, are large and require large sheaths to accommodate and help deliver them to the desired site within the body.

BRIEF SUMMARY OF THE INVENTION

[0004] In accordance with an example hereof, an introducer comprises: an introducer hub, the introducer hub including a proximal end, a distal end, and a central lumen extending from the proximal end to the distal end, the introducer hub including a distal tube at a distal portion of the introducer hub, the distal tube defining a portion of the central lumen; a sheath coupled to the introducer hub and extending distally therefrom; and a connector coupling the sheath to the distal tube of the introducer hub.

[0005] In accordance with another example hereof, in the introducer of any of the preceding or following examples: the distal tube comprises a cylinder having a stepped outer surface and including tube threads on a portion of the stepped outer surface; the connector comprises a cap comprising a cylinder having a stepped inner surface and including cap threads on a portion of the stepped inner surface; and the cap is configured to capture a proximal end of the sheath between the stepped inner surface of the cap and the stepped outer surface of the distal tube as the cap threads engage with the tube threads.

[0006] In accordance with another example hereof, in the introducer of any of the preceding or following examples: the distal tube further comprises a ratchet lock proximal of the tube threads, the ratchet lock comprising a plurality of teeth extending radially outwardly and circumferentially around the distal tube; and the cap further comprises at least one tooth extending radially inwardly proximal of the cap threads, wherein the at least one tooth is configured to engage the plurality of teeth of the ratchet lock to lock the cap to the distal tube.

[0007] In accordance with another example hereof, in the introducer of any of the preceding or following examples: the distal tube comprises a tube lip extending radially outwardly proximal of

the tube threads; the cap includes a plurality of longitudinal legs extending proximally such that a proximal end of the legs is proximal of the cap threads, the plurality of longitudinal legs disposed adjacently around a circumference of the cap and including gaps between circumferentially adjacent longitudinal legs, the plurality of longitudinal legs comprising a cap lips extending radially inwards from the proximal end of the longitudinal legs; and the plurality of longitudinal legs are configured to flex radially outwardly for the cap lips to extend over the tube lip and are configured to flex back radially inwardly when the leg lips extend proximal of the tube lip such that the leg lips lock the cap to the distal tube.

[0008] In accordance with another example hereof, in the introducer of any of the preceding or following examples, the connector comprises a housing, a grab ring, and an O-ring.

[0009] In accordance with another example hereof, in the introducer of any of the preceding or following examples: the housing comprises a tube including distal lip extending radially inwardly from a distal end of the housing, a plurality of longitudinal legs at a proximal end of tube, the longitudinal legs separated circumferentially by a plurality of gaps between adjacent legs of plurality of longitudinal legs, and a proximal lip extending radially inward from a proximal end of the longitudinal legs; the distal tube includes a first groove extending radially inwardly from an outer surface of the distal tube and circumferentially around the distal tube and a second groove extending radially outwardly from an inner surface of the distal tube and circumferentially around the inner surface of the distal tube; and with a proximal end of the sheath disposed within the distal tube, the O-ring is disposed in the second groove of the distal tube between an inner surface of the distal tube and an outer surface of the sheath, the housing is configured to be translated over the sheath and over the distal tube proximally such that the longitudinal legs flex radially outwardly as the proximal lip extend over the distal tube until the proximal lip reaches the first groove such that the longitudinal legs flex back radially inwardly as the proximal lip is disposed within the first groove, and the distal lip of the housing extends radially inwardly distal of the distal end of the distal tube and abuts the outer surface of the sheath.

[0010] In accordance with another example hereof, in the introducer of any of the preceding or following examples, the grab ring is disposed between the distal lip of the housing and the distal end of the distal tube.

[0011] In accordance with another example hereof, in the introducer of any of the preceding or following examples: the connector comprises a locking clip configured to be disposed over the sheath, wherein the locking clip comprises a generally cylindrical tube including at least one clip arm proximal of a distal end of the locking clip, wherein the at least one clip arm is configured to flex radially inwardly and outwardly and includes a clip lip extending radially outwardly from the clip arm; the distal tube includes at least one slot extending from an outer surface to an inner surface thereof; the sheath includes a proximal lip extending radially outwardly therefrom; and with the locking clip disposed over the sheath such that a proximal end of the locking clip is adjacent the proximal lip of the sheath, the sheath and locking clip are configured to be inserted into the distal tube such that the at least one clip arm flexes radially inwardly until the clip lip aligns with the slot of the distal tube such the at least one clip arm flexes back radially outwardly and the clip lip extends into the slot to couple the sheath and the locking clip to the distal tube.

[0012] In accordance with another example hereof, in the introducer of any of the preceding or following examples, the locking clip further includes a distal lip extending radially outwardly from the distal end of the locking clip, wherein the distal lip is configured to abut a distal end of the distal tube when the clip lip extends into the slot.

[0013] In accordance with another example hereof, in the introducer of any of the preceding or following examples, the connector comprises a generally cylindrical tube including a distal portion overmolded over a proximal end of the sheath, a proximal portion configured to be disposed over the distal tube of the introducer hub, and band configured to be disposed around the proximal portion to couple the proximal portion to the distal tube of the introducer hub.

[0014] In accordance with another example hereof, in the introducer of any of the preceding or following examples, the distal tube of the introducer hub includes barbs extending radially outwardly along the distal tube, wherein the band of the connector is configured to deform the proximal portion of the generally cylindrical tube of the connector to couple the generally cylindrical tube of the connector to the distal tube of the introducer hub.

[0015] In accordance with another example hereof, in the introducer of any of the preceding or following examples, each barb extends circumferentially around the distal tube, and wherein each barb includes a barb distal end, a ramp extending radially outwardly and proximally from the barb distal end to a barb proximal end, and a shoulder at the barb proximal end extending radially inwardly from the barb proximal end.

[0016] In accordance with another example hereof, a system comprises: a dilator comprising a dilator hub and a dilator shaft coupled to and extending distally from the dilator hub, the dilator shaft including: a proximal portion having a first outer diameter; a middle portion having a second outer diameter; a proximal transition portion disposed between the proximal portion and the middle portion, the proximal transition portion tapering in a distal direction from the first outer diameter to the second outer diameter; and a distal portion extending distally from the middle portion, the distal portion including a distal cylindrical portion having a third outer diameter, wherein the first outer diameter is larger than the second outer diameter and the third outer diameter, and wherein the third outer diameter is larger than the second outer diameter.

[0017] In accordance with another example hereof, in the system of any of the preceding or following examples, the dilator shaft further includes a distal transition portion disposed between the middle portion and the distal cylindrical portion, wherein the distal transition portion flares outwardly distally from the second outer diameter of the middle portion to the third outer diameter of the distal cylindrical portion.

[0018] In accordance with another example hereof, in the system of any of the preceding or following examples, the distal transition portion flares linearly outwardly from a proximal end adjacent a distal end of the middle portion to a distal end of the distal transition portion adjacent a proximal end of the distal cylindrical portion.

[0019] In accordance with another example hereof, in the system of any of the preceding or following examples, the distal transition portion includes a proximal section and a distal section, the proximal section flaring radially outwardly with a curved concave profile, and a distal section flaring outwardly with a curved convex profile.

[0020] In accordance with another example hereof, in the system of any of the preceding or following examples, the distal transition portion includes a proximal section extending distally from the middle portion, a middle section extending distally from the proximal section, and a distal section distal extending distally from the middle section, wherein the proximal section flares outwardly in a distal direction, the middle section is generally cylindrical, and the distal section flares outwardly in a distal direction.

[0021] In accordance with another example hereof, in the system of any of the preceding or following examples, the proximal section and the distal section flare linearly outwardly.

[0022] In accordance with another example hereof, in the system of any of the preceding or following examples, each of the proximal section and the distal section includes curved concave profile and a curved convex profile.

[0023] In accordance with another example hereof, in the system of any of the preceding or following examples, the distal portion of the dilator shaft further includes a distal tapered tip extending distally from the distal cylindrical portion wherein the distal tapered tip includes an outer diameter that tapers distally.

[0024] In accordance with another example hereof, in the system of any of the preceding or following examples, the system further comprises an introducer comprising an introducer hub and an expandable sheath coupled to and extending distally from the introducer hub, wherein the

expandable sheath is expandable from a radially compressed state wherein the expandable sheath has a first inner sheath diameter and a first outer sheath diameter to a radially expanded state wherein the sheath has a second inner sheath diameter larger than the first inner sheath diameter and a second outer sheath diameter larger than the first outer sheath diameter.

[0025] In accordance with another example hereof, in the system of any of the preceding or following examples: the first outer diameter of the dilator shaft proximal portion is sized to fit within the introducer hub; the second outer diameter of the dilator shaft middle portion is smaller than the first inner sheath diameter of the sheath; and the third outer diameter of the dilator shaft distal cylindrical portion is approximately equal to the first outer sheath diameter of the sheath.

[0026] In accordance with another example hereof, an expandable introducer sheath comprises: a shaft configured to be expanded from a radially unexpanded, folded state to a radially expanded, unfolded state in reaction to a device being extended through a central lumen of the expandable sheath, wherein the shaft includes: an inner liner; and an outer wall disposed radially outwardly of the inner liner, the outer wall having a variable thickness around a circumference of the shaft; and a tip coupled to a distal portion of the shaft, the tip including a first circumferential portion and a second circumferential portion, wherein the second circumferential portion is more flexible than the first circumferential such that the second circumferential portion is configured to radially expand when the in reaction to the device extending through tip.

[0027] In accordance with another example hereof, in the expandable introducer sheath of any of the preceding or following examples, in the radially expanded, unfolded state, the shaft includes a thick-walled region and a thin-walled region, wherein the thin-walled region is configured to fold when the shaft is in the radially unexpanded, folded state.

[0028] In accordance with another example hereof, in the expandable introducer sheath of any of the preceding or following examples, in the radially unexpanded, folded state, the shaft includes a folded region and an unfolded region, wherein the second circumferential portion is circumferentially aligned with the folded region.

[0029] In accordance with another example hereof, a method of forming a tip of an introducer sheath comprises: loading a shaft over a mandrel, the shaft including a liner, an inner layer, and an outer layer on a mandrel, the liner being disposed on the mandrel, the inner layer being disposed over the liner, and the outer layer being disposed over the inner layer, wherein the outer layer does not extend around the entire circumference of the inner layer, and wherein the liner, the inner layer, and the outer layer are not fused together; inserting a shim between the outer layer and the inner layer at a tip region of the shaft; fusing the liner, the inner layer, and the outer layer along the shaft except where the shim is disposed between the inner layer and the outer layer; removing the shim from between the inner layer and the outer layer, thereby leaving a gap between the inner layer and the outer layer in the distal region; folding the shaft to a radially unexpanded, folded state by folding a foldable region of the shaft where the outer layer does not extend around the circumference of the inner layer; shape setting the shaft to the radially unexpanded, folded state; removing an inner fold of the folded portion in the tip region; inserting a tip extrusion between the inner layer and the outer layer within the shaft in the radially unexpanded, folded shape, the tip extrusion including a first circumferential portion and a second circumferential portion, wherein the second circumferential portion is more flexible than the first circumferential portion, wherein the second circumferential portion is circumferentially aligned with the folded portion of the shaft; fusing the tip region of the shaft such that the first circumferential portion of the tip extrusion is fused to the inner layer and the outer layer, and the second circumferential portion of the tip extrusion is not fused to the inner layer and the outer layer.

[0030] In accordance with another example hereof, the method of any of the preceding or following examples further comprises, prior to fusing the tip region, inserting a tip extrusion shim under the second circumferential portion of the tip extrusion to prevent the second circumferential portion from fusing with the inner layer and the outer layer.

[0031] The details of one or more aspects of the disclosure are set forth in the accompanying drawings and the description below. Other features, objects, and advantages of the techniques described in this disclosure will be apparent from the description and drawings, and from the claims.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0032] The foregoing and other features and advantages of the present disclosure will be apparent from the following description of embodiments hereof as illustrated in the accompanying drawings. The accompanying drawings, which are incorporated herein and form a part of the specification, further serve to explain the principles of the present disclosure and to enable a person skilled in the pertinent art to make and use the embodiments of the present disclosure. The drawings may not be to scale.

[0033] FIG. 1 shows a perspective side view of an expandable introducer according to embodiments hereof.

[0034] FIG. 2 is an exploded view of the expandable introducer of FIG. 1.

[0035] FIG. 3 shows a cross-section of a sheath of the expandable introducer of FIG. 1 in an expanded, unfolded state according to embodiments hereof.

[0036] FIG. 4 shows a cross-section of a sheath of the expandable introducer of FIG. 1 in an unexpanded, folded state according to embodiments hereof.

[0037] FIG. 5 shows a cross-section of a sheath of the expandable introducer of FIG. 1 in an expanded, unfolded state according to embodiments hereof.

[0038] FIG. 6 shows perspective view of a portion of a coil of the sheath of the expandable introducer of FIG. 1.

[0039] FIG. 7A shows a schematic illustration of a hybrid flexible tip for forming the tip of the expandable introducer of FIG. 1.

[0040] FIG. 7B shows a schematic cross-sectional view taken along line 7B-7B of FIG. 7A according to embodiment hereof.

[0041] FIG. 7C shows a schematic cross-sectional view taken along line 7B-7B of FIG. 7A according to another embodiment hereof.

[0042] FIGS. 8-14 show steps in a method of forming a tip of the expandable introducer of FIG. 1 according to embodiments hereof.

[0043] FIG. 15 shows a perspective distal end view of the tip of the expandable introducer of FIG. 1 according to embodiments hereof.

[0044] FIG. 16 shows a side view of the tip of the expandable introducer of FIG. 1 with a delivery catheter disposed through the tip according to embodiments hereof.

[0045] FIG. 17 shows an introducer hub of an introducer according to embodiments hereof.

[0046] FIG. 18 shows a cap of an introducer configured to couple an introducer sheath to the introducer hub of FIG. 17.

[0047] FIG. 19 shows an introducer with the cap of the FIG. 18 coupling an introducer sheath to the introducer hub of FIG. 17.

[0048] FIG. 20 shows a close-up view of a portion of the introducer hub of FIG. 17.

[0049] FIG. 21 shows longitudinal cross-sectional view of a sheath coupled to the introducer hub of FIG. 17 using the cap of FIG. 18.

[0050] FIGS. 22-25 show steps in a method of coupling an introducer sheath to the introducer hub of FIG. 17 using the cap of FIG. 18.

[0051] FIG. 26 shows a longitudinal cross-section of an introducer according to embodiments hereof.

[0052] FIG. 27 shows a close-up view of a proximal end of a cap of the introducer of FIG. 26 locking with an introducer hub of the introducer of FIG. 26.

[0053] FIGS. 28 and 29 show side views of steps in a method of coupling an introducer sheath of the introducer of FIG. 26 using the cap of FIG. 26.

[0054] FIG. 30 shows an O-ring of a connector for connecting a sheath to an introducer hub of an introducer according to embodiments hereof.

[0055] FIG. 31 shows a grab ring of a connector for connecting a sheath to an introducer hub of an introducer according to embodiments hereof.

[0056] FIG. 32 shows a housing of a connector for connecting a sheath to an introducer hub of an introducer according to embodiments hereof.

[0057] FIG. 33 shows the grab ring of FIG. 31 disposed within the housing of FIG. 32.

[0058] FIG. 34 shows a side view of an introducer with the connector of FIGS. 31-33 coupling a sheath of the introducer to an introducer hub of the introducer.

[0059] FIG. 35 is a longitudinal cross-sectional view of the introducer of FIG. 34 with the connector of FIGS. 31-33 coupling the sheath to the introducer hub.

[0060] FIG. 36 shows an exploded view of an introducer according to embodiments hereof.

[0061] FIG. 37 shows a side view of the introducer of FIG. 36 with a locking clip thereof disposed on a sheath thereof.

[0062] FIG. 38 shows the introducer of FIG. 36 with the locking clip thereof coupling the sheath to the introducer hub.

[0063] FIG. 39 shows a perspective view of an introducer according to embodiments hereof.

[0064] FIG. 40 shows a longitudinal cross-sectional view of the introducer of FIG. 39.

[0065] FIG. 41 shows a perspective longitudinal cross-sectional view of the introducer of FIG. 39.

[0066] FIG. 42A shows a schematic illustration of a dilator according to embodiments hereof.

[0067] FIG. 42B shows a close-up view of a distal portion of the dilator shaft of the dilator of FIG. 42A.

[0068] FIG. 43 shows the dilator of FIG. 42A and an expandable introducer according to embodiments hereof.

[0069] FIG. 44 shows a distal portion of the dilator shaft of the dilator of FIG. 42A disposed through an introducer sheath of an introducer.

[0070] FIG. 45 shows a cut-away view of an introducer hub, a hemostatic sleeve, an introducer sheath, and a portion of the dilator shaft of FIG. 42A.

[0071] FIG. 46 shows a distal portion of the dilator shaft of the dilator of FIG. 42A disposed through an introducer sheath of an introducer.

[0072] FIG. 47 shows a distal portion of the dilator shaft of the dilator of FIG. 42A disposed through an introducer sheath of an introducer.

[0073] FIGS. 48-50 show the dilator shaft of the dilator of FIG. 42A being retracted proximally through an expandable introducer sheath.

[0074] FIG. 51 depicts a side view of a distal transition portion of the dilator of FIG. 42B, according to embodiments hereof.

[0075] FIG. 52 depicts a partial side view illustration of a distal tip of a sheath and the distal transition portion of the dilator of FIG. 51 according to embodiments hereof.

[0076] FIG. 53 depicts a side view of a distal transition portion of the dilator of FIG. 42B according to embodiments hereof.

[0077] FIG. 54 depicts a partial side view illustration of a distal tip of a sheath and the distal transition portion of the dilator of FIG. 53 according to embodiments hereof.

[0078] FIG. 55 depicts a side view of a distal transition portion of the dilator of FIG. 42B according to embodiments hereof.

[0079] FIG. 56 depicts a partial side view illustration of a distal tip of a sheath and the distal transition portion of the dilator of FIG. 55 according to embodiments hereof.

[0080] FIG. 57 depicts a side view of a distal transition portion of the dilator of FIG. 42B according to embodiments hereof.

[0081] FIG. 58 depicts a partial side view illustration of a distal tip of a sheath and the distal transition portion of the dilator of FIG. 57 according to embodiments hereof.

DETAILED DESCRIPTION

[0082] It should be understood that various embodiments disclosed herein may be combined in different combinations than the combinations specifically presented in the description and accompanying drawings. It should also be understood that, depending on the example, certain acts or events of any of the processes or methods described herein may be performed in a different sequence, may be added, merged, or left out altogether (e.g., all described acts or events may not be necessary to carry out the techniques). In addition, while certain aspects of this disclosure are described as being performed by a single device or component for purposes of clarity, it should be understood that the techniques of this disclosure may be performed by a combination of devices or components associated with, for example, a delivery device. The following detailed description is merely exemplary in nature and is not intended to limit the invention of the application and uses of the invention. Furthermore, there is no intention to be bound by any expressed or implied theory presented in the preceding field of the invention, background, summary, or the following detailed description.

[0083] As used in this specification, the singular forms “a”, “an” and “the” specifically also encompass the plural forms of the terms to which they refer, unless the content clearly dictates otherwise. The term “about” is used herein to mean approximately, in the region of, roughly, or around. When the term “about” is used in conjunction with a numerical range, it modifies that range by extending the boundaries above and below the numerical values set forth. In general, the term “about” is used herein to modify a numerical value above and below the stated value by a variance of 20%. It should be understood that use of the term “about” also includes the specifically recited number of value.

[0084] The terms “proximal” and “distal” herein are used with reference to the clinician using the devices. Therefore, “proximal” and “proximally” mean in the direction toward the clinician, and “distal” and “distally” mean in the direction away from the clinician. In other words, “proximal” and “proximally” mean towards the hub end of the introducer and “distal” and “distally” means toward the tip end of the introducer.

[0085] FIG. 1 shows an expandable introducer **100** according to embodiments herein. The expandable introducer **100** includes a proximal end **102** and a distal end **104**. The expandable introducer **100** further includes an expandable sheath **110**, a distal tip **130**, a hemostatic sleeve **140**, an introducer hub **150**, and an extension tube **170**. FIG. 2 shows an exploded view of the expandable introducer **100**. As shown generally in FIG. 2, the expandable sheath **110** includes a shaft **112** and a coil **114**, as described in more detail below. The distal tip **130** is coupled to and covers a distal end of the expandable sheath **110**. A proximal end of the expandable sheath **110** is coupled to a distal end of the introducer hub **150**. The hemostatic sleeve **140** is disposed over a proximal portion of the expandable sheath **110** and is coupled to the introducer hub **150**. A proximal portion of the expandable sheath **110** is coupled to the introducer hub **150**, as explained in more detail below.

[0086] The expandable introducer **100** may further include an extension tube **170** coupled to the introducer hub **150**. In particular, a proximal end of the extension tube **170** may be coupled to a side port is coupled to a side port **152** of the introducer hub **150**. The side port **152** is fluidly coupled to a passageway of the introducer hub **150**, which is fluidly coupled to a central lumen of the sheath **110**. Therefore, injecting saline, or other fluids, into the extension tube **170** results in the saline or other fluid being injected into the central lumen of the sheath **110**. A stopcock **172** is coupled to the extension tube **170** and is configured to control the flow of saline, or other fluids, through the extension tube **170** and into the vasculature of the patient.

[0087] The introducer hub **150** of FIGS. **1** and **2** may be similar to the introducer hub described in International Patent Publication No. WO 2023/147422, which is incorporated by reference herein in its entirety. Thus, the introducer hub **150** generally includes a hub body **154**, seal section **160**, and a seal cap **168**. The hub body **154** includes a lumen or passageway **156** extending from a proximal end to a distal end thereof. The seal section **160** may include a first or proximal seal **162**, a second or middle seal **164**, and a third or distal seal **166**, as shown in FIG. **2**. However, this is not meant to be limiting. For example, the seal section **160** may include exactly one seal, or exactly two seals, or exactly four seals, or four or more seals. In the embodiment shown, the first seal **162** is the proximal-most seal and the third seal **166** is the distal-most seal. The second seal **164** is disposed between, and is coupled to, the first seal **162** and the third seal **166**. It should be understood that the first seal **162**, the second seal **164**, and the third seal **166** maintain hemostasis when various devices are inserted through the seals and when all devices are removed from the seals. The first seal **162** may also be referred to as a catheter seal, the second seal **164** may also be referred to as a slit seal, and the third seal **166** may also be referred to as a guidewire seal. The seal section **160** and the seals **162**, **164**, **166** may be similar to the seals described in International Patent Publication No. WO 2023/147422, which is incorporated by reference herein in its entirety. The seal cap **168** may be a generally cylindrically shaped piece of the introducer hub **150** that includes a body and a central lumen or passageway extending longitudinally therethrough. The seal cap **168** is configured to mate with the proximal end of the hub body **154** and abut the seal section **160** disposed within the hub body **154**, maintaining compression in the seals and securing the seals in place. The seal cap **168** may be similar to the seal caps described in International Patent Publication No. WO 2023/147422, which is incorporated by reference herein in its entirety.

[0088] FIGS. **3-5** show cross-sectional views of the sheath **110** according to embodiments herein. FIGS. **3-5** show the sheath **110** in an unexpanded, folded state and the FIG. **6** show the sheath in an expanded, unfolded state. As shown in FIG. **2**, the sheath **110** includes the shaft **112** and the coil **114**. As shown in more detail in FIGS. **3-5**, the sheath **110** includes a liner **116** and an outer wall **117** disposed radially outside and coupled to an outer surface of the liner **116**. The coil **114** is embedded within the outer wall **117**, as described in more detail below.

[0089] In embodiments, the liner **116** may be a thin, difficult to pierce, low-friction plastic tube disposed within and coupled to the interior surface of the outer wall **117**. The liner **116** is configured to reduce friction forces and aid in the translation of devices through the central lumen of the sheath **110**. In an embodiment, the liner **116** may have a thickness of about 0.0002 inch that is constant, or continuous, around the circumference of the sheath **110**. The liner **116** may be polytetrafluoroethylene (PTFE), fluorinated ethylene propylene (FEP) or other materials known to those skilled in the art suitable for the purposes described herein, or any combination thereof.

[0090] In embodiments, the outer wall **117** is disposed outside of the liner **116** and may comprise polyether block amine (PEBAX) with a Shore D durometer value of about 63 D to about 72 D. In the embodiments shown, the outer wall **117** has a varying thickness around the circumference of the sheath **110**, as described below.

[0091] In embodiments, the coil **114** is formed of a wire comprising a shape memory material, such as nitinol. In embodiments, the wire has a diameter of about 0.008 inch. As shown in FIG. **3**, in the radially expanded, unfolded state, the coil **114** extends partially around the circumference of the sheath **110**. As can be seen in FIG. **4**, in the radially unexpanded, folded state, the circumferential ends **115A**, **115B** of the coil **114** are closer to each other than in the radially expanded, unfolded state. The coil **114** is shape set to the radially unexpanded state shown in FIG. **3** such that after a device is passed through the sheath **110** to radially expand the sheath **110**, the coil **114** will return the sheath **110** to the radially unexpanded, folded state of FIG. **4**. As shown in FIG. **6**, the coil **114** may comprise a series of wrap or circumferential portions **123** and a series of crowns or bends **121**, as described, for example, in U.S. Published Application Publication No. 2014/0236122 and/or U.S. Provisional Application No. 63/510,670, filed Jun. 28, 2023, both of which are incorporated

by reference herein in their entirety. In the embodiment of FIGS. 3-6, the circumferential ends **115A**, **115B** of the coil **114** do not overlap in the radially unexpanded, folded configuration, as shown in FIGS. 4 and 6. As can be seen in FIG. 6, the crowns or bends **121** define the circumferential ends **115A**, **115B** of the coil.

[0092] In the radially expanded, unfolded state, the sheath **110** may have an inner diameter **D1**, as shown in FIGS. 3 and 6. In embodiments, the inner diameter **D1** may be in the range of about 17 French (about 5.67 mm) to about 22 French (about 7.33 mm). However, this is not meant to be limiting, and the inner diameter **D1** may be any diameter suitable for the purposes described herein. In embodiments, the inner diameter **D2** with the sheath **110** in the radially unexpanded, folded state shown in FIG. 5 may be in the range of about 10 French to about 16 French (about 3.33 mm to about 5.33 mm) depending on the application.

[0093] As shown in FIG. 4, the sheath **110** in the radially unexpanded, folded state includes a folded region **122** and a non-folded region **124**. Also, in the radially expanded, unfolded state, the sheath **110** includes a coiled region **126** and a non-coiled region **128**, as shown in FIG. 3, which may also be referred to as a thick-walled region **126** and a thin-walled region **128**. In embodiments, the thick-walled region **126** has a first wall thickness **T1** larger than a second wall thickness **T2** of the thin-walled region **128**. As shown in FIG. 3, the non-coiled or thin-walled region **128** includes the liner **116** and a thin portion of the outer wall **117**. The coiled or thick-walled region **128** includes the liner **116**, a thick portion of the outer wall **117**, and the coil **114** embedded within the outer wall **117**. Thus, the sheath **110** has variable wall thickness. In embodiments, the thickness of the outer layer **120** may taper to a thinner thickness as the sheath **110** transitions from the thick-walled region **126** to the thin-walled region **128**. The second thickness **T2** in the thin-walled region **128** may be about 0.0065 inch and the first thickness **T1** in the thick-walled region **126** may be about 0.0225 inch. However, this is merely an example and is not meant to be limiting. The reduced wall thickness in the thin-walled/non-coiled region **128** assists in enabling the sheath **110** to fold into the radially unexpanded, folded state shown in FIG. 4.

[0094] As can be seen in FIG. 4, in the radially unexpanded, folded state, the coil **114** extends into the folded region **122** such that the bends/crowns **121** (circumferential ends **115A/115B**) of the coil **114** are disposed adjacent to each other. As can be seen in FIG. 3, in the radially expanded, unfolded state, the bends/crown **121** (circumferential ends **115A/115B**) of the coil **114** are spaced from each other, thereby forming the uncoiled region **128**, or a gap, between the circumferential ends **115A**, **115B** of the coil **114**. In an embodiment, the coil **114** extends greater than 50% around the circumference of the sheath **110** in the radially expanded, unfolded state, as shown in FIG. 3.

[0095] The sheath **110** is configured to radially expand from the radially unexpanded, folded state to the radially expanded, unfolded state, and radially compress back to the radially unexpanded, folded state. The sheath **110** maintains the radially unexpanded, folded state until a device is advanced through the sheath **110**, during which the sheath **110** is configured to expand to the radially expanded, unfolded state. When the sheath **110** is in the radially unexpanded, folded state, the sheath **110** includes the folded region **122** and the non-folded region **124**, as shown in FIG. 4. The folded region **122** is defined as the region where a section of the sheath **110** is folded over, as can be seen in FIG. 4. In other words, a portion of the sheath **110** is folded and bent to overlap with itself. The non-folded region **124** of the sheath **110** is defined as the remainder of the sheath **110** that does not include a fold. As can be seen in FIG. 4, when folded, the sheath **110** includes two folds or bends **113**.

[0096] The liner **116** and the outer wall **117** may be formed by co-extrusion or a layered reflow lamination processes. In particular, in an embodiment, as shown in FIG. 5, the outer wall **117** may be formed of two layers of the same material. In particular, an inner layer **118** and an outer layer **120** of the same material may form the outer wall **117**. In an embodiment, prior to forming the sheath **110**, as shown in FIG. 5, the inner layer **118** is continuous around the entire circumference of the liner **116** and the outer layer **120** is not continuous and extends only partially around the

circumference of the inner layer **118**. Further, prior to forming the sheath **110**, the coil **114** is disposed between the inner layer **118** and the outer layer **120**. In an embodiment, the inner layer **118** may have thickness of about 0.0045 inch and the outer layer **120** may have a thickness of about 0.0016 inch. In such an embodiment, both inner layer **118** and the outer layer may comprise polyether block amine (PEBAX) with a Shore D durometer value of about 63 D to about 72 D. In the embodiments, the inner layer **118** and the outer layer **120** are formed from the same material having the same durometer. In particular, in an embodiment, the inner layer **118** and the outer layer **120** may both be formed of PEBAX with the same Shore D durometer and the durometer being about 63 D. Thus, upon reflow, the inner layer **118** and the outer layer **120** meld together to form a single outer wall **117** with the coil **114** embedded therein.

[0097] As briefly explained above, the tip **130** is coupled to and covers a distal end of the sheath **110**. FIGS. 7A and 7B show a tip extrusion **132** and FIGS. 8-14 show a method of forming the tip **130** according to embodiments hereof. The coil **114** does not extend to the distal end of the sheath **110**. Accordingly, there is a small length of the sheath **110** at the distal end thereof that does not include the coil **114**. This distal end of the sheath **110** without the coil **114** is coupled to the tip extrusion **132** to form the tip **130**.

[0098] The tip extrusion **132** is shown in FIGS. 7A-7C. As shown in FIG. 7A, the tip extrusion **132** is a generally cylindrical tube. In an embodiment, the tip extrusion has a length of about 6 mm to about 9 mm. FIG. 7B is a cross-section view of the tip extrusion **132** taken along line 7B-7B of FIG. 7A. The tip extrusion **132** includes a first circumferential portion **134** made of a first material and a second circumferential portion **136** made of a second material. The first material may be stiffer than the second material. The second material is flexible such that the second circumferential portion **136** is flexible and can expand with the folded portion of the sheath **110** when the sheath **110** radially expands. Thus, the first circumferential portion **134** is relatively stiff to aid in pushability of the distal tip **130** during insertion into a vessel, and the second circumferential portion **136** is relatively flexible to enable the distal tip **130** to conform to the profile of the inserted device, i.e., radially expand to accommodate the inserted device. In an embodiment, the first material is PEBAX/polyamide, and the second material is polyurethane/soft durometer PEBAX.

[0099] FIG. 7C shows another embodiment of a cross-section of the tip extrusion **132**. In the embodiment of FIG. 7C the tip extrusion **132** includes three layers **137**, **138**, **139**. In the embodiment shown, a first layer **137** extends 360 degrees. Further, the first layer **137** is flexible. The second layer **138** and the third layer **139** sandwich the first layer **137** and extend only partially around the inner and outer circumference thereof, respectively. Thus, the portion of the circumference with the first, second, and third layers **137**, **138**, **139** forms the first circumferential portion **134**, and the portion of the circumference with only the first layer **137** forms the second circumferential portion **136**. In an embodiment, the first layer **137** may be polyurethane, soft durometer PEBAX and the second and third layers **138**, **139** may be PEBAX, polyamide. In other words, the first layer **137** is a soft, flexible material and the second and third layers **138**, **139** are stiffer materials relative to the first layer **137**. Further, although shown as layers, those skilled in the art will recognize that the layers are fused to form a single wall. Further, although shown with three layers with the middle layer being the flexible layer, those skilled in the art will recognize that other embodiments may be used, such as, but not limited to, two layers, or four or more layers, and that the flexible layer need not be the middle layer.

[0100] In a step of the method, the liner **116**, the inner layer **118**, and the outer layer **120** are loaded onto a straight mandrel **190**, as shown in FIG. 8. The liner **116**, the inner layer **118**, and the outer layer **120** are not bonded together at this stage of the method. As explained above, the outer layer **120** does not extend around the entire circumference of the sheath **110**. Therefore, in FIG. 7, a portion of the circumference of the sheath **110**, i.e., the thin-walled region **126**, does not include the outer layer **120** such that the inner layer **118** is the outermost layer of the thin-walled region **126**.

[0101] In another step of the method, a shim **192** is inserted between the outer layer **120** and the

inner layer **118**, i.e., under the outer layer **120** and over the inner layer **118**, as shown in FIG. **9**. The shim **192** may be a cylindrical tube approximately 6 mm in length. As shown in FIG. **9**, only a portion of the circumference of the shim **192** is shown because a portion of the circumference of the shim **192** is disposed under the outer layer **120**. The portion of the circumference of the shim **192** that is shown in FIG. **9** is aligned with the thin-walled region **126** of the circumference of the sheath **110**. Because the thin-walled region **126** does not include an outer layer **120**, the shim **192** is not hidden under the outer layer **120** in the thin-walled region **126** in FIG. **9**.

[0102] With the shim **192** inserted between the inner layer **118** and the outer layer **120**, a heat shrink wrap is loaded over the assembly. Heat and pressure are applied to fuse the layers together except in the area of the shim **192**. The shim **192** may be formed of a material such as PTFE. During this step, the inner layer **118** (PEBAX) fuses to the liner **116** (PTFE) because the liner **116** is etched to allow for adherence. The fuse may be described as thermal tack rather than a weld, but it provides sufficient adherence for the device to function as desired. The shim **192** (PTFE) is not etched and therefore does not adhere to the inner layer **118** (PEBAX) or the outer layer **120** (PEBAX), and prevents fusion of the inner layer **119** to the outer layer **120**.

[0103] The shim **192** is then removed from under the outer layer **120**, as shown in FIG. **10**, thereby leaving a gap between the inner layer **118** and the outer layer **120** at the distal end of the sheath **110** (i.e., the gap is equal to the length of the shim **192** inserted between the inner layer **118** and the outer layer **120**). The sheath **110** is then folded to the radially unexpanded, folded state. A heat shrink wrap is loaded over the full length of the sheath **110**. Heat and pressure are applied to set the shape of the sheath **110** in the radially unexpanded, folded state and the unfused layers are fused.

[0104] In another step, the inner fold **198** of the folded portion is cut away for a short distance from the distal end, as shown in FIG. **11B**. In an example, about 3 mm of the distal end of the inner fold **198** of the folded portion is cut away. Due to this cut, after reflow, the distal profile of the tip will be circular, as shown in FIG. **15**. The tip with the inner fold **198** cut away is shown in FIG. **11A**.

[0105] In another step, the tip extrusion **132** is loaded under the outer layer **120** and over the inner layer **118**, as shown in FIG. **12**. The second (flexible) circumferential portion **136** of the tip extrusion **132** is circumferentially aligned with the folded portion of the sheath **110**. In other words, the second (flexible) circumferential portion **136** of the tip extrusion **132** is circumferentially aligned between the circumferential ends **119A**, **119B** of the outer layer **120**. Thus, the first (stiff) circumferential portion **134** of the tip extrusion **132** is circumferentially aligned with the outer layer **120**. Further, a shim **194** is inserted under the second (flexible) circumferential portion **136** of the tip extrusion **132** and over the liner **116**, as shown in FIG. **12**. As can be seen in FIG. **12**, when the shim **194** is inserted under the tip extrusion **132**, the shim **194** forms an arc or partial cylinder shape. In practice, the shim **194** may be relatively flat and may bend to conform to the shape of the second (flexible) circumferential portion **136** of the tip extrusion **132**, as shown in FIG. **12**. The shim **194** prevents the second (flexible) circumferential portion **136** of the tip extrusion **132** from bonding to other layers during fuse in the folded region **122** of the sheath **110**.

[0106] In another step of the method, a heat shrink wrap is loaded over the tip assembly. Heat and pressure are applied to fuse the materials of the tip assembly together, except where the shim **194** is located. Thus, during fusing, the tip extrusion **132** is fused to inner and outer layers **118**, **120** in the non-folded region **124** of the sheath **110**. The shim **194** prevents the tip extrusion **132** from bonding to the inner and outer layers **118**, **120** in the folded region **122** of the sheath **110**. After fusing, the shim **194** is removed and excess liner **116** is trimmed.

[0107] Upon completion of the method, the second (flexible) circumferential portion **136** of the tip extrusion **132** is circumferentially aligned with the folded region **122** of the sheath **110** and the first (stiff) circumferential portion **134** of the tip extrusion **132** is circumferentially aligned with the non-folded region **124** of the sheath **110**. FIG. **13** shows the second (flexible) circumferential portion **136** and FIG. **14** shows the first (stiff) circumferential portion **134**. As explained above, FIG. **15**

shows the circular distal profile of the tip **130**. FIG. **16** shows a delivery catheter **196** inserted through the distal tip **130**, showing the second (flexible) circumferential portion **136** of the tip extrusion **132** radially expanding to accommodate the delivery catheter **196**.

[0108] FIGS. **17-25** show devices and methods for securing a sheath of an introducer to an introducer hub **200** of the introducer. In embodiments, the sheath may be the sheath **110** as described above or may be other sheaths. In embodiments, the introducer hub **200** may be similar to the introducer hub **150** described above except for details described herein with respect to the introducer hub **200**. However, this is not meant to be limiting, and the introducer hub **200** is not limited to the details described above with respect to the introducer hub **150**.

[0109] FIG. **17** shows the introducer hub **200** according to embodiments herein. Those skilled in the art will recognize that all the details of the introducer hub **200** are not described and that other features may be included with the introducer hub **200** or that features may be excluded or modified. The introducer hub **200** includes a body **202** having a proximal end **204**, a distal end **206**, and a central lumen or passageway **208** extending from the proximal end **204** to the distal end **206**. The introducer hub **200** may include seals, seal caps, and other features as described above or as known to those skilled in the art. The introducer hub **200** further includes a cap **250** for coupling the sheath **110** to the hub body **202**.

[0110] In the embodiment shown, the body **202** of the introducer hub **200** includes a distal tube **210** extending proximally from the distal end **206** of the body. The distal tube **210** includes a first portion **212**, a second portion **214** proximal of the first portion **212**, and a ratchet lock **216** proximal of the second portion **214**. In particular, as shown in FIGS. **17**, **20**, and **21**, the first portion **212** is a cylindrical tube with an inner surface thereof defining a portion of the central passageway **208**. The wall of the first portion has a first wall thickness such that an outer surface thereof defines a first outer diameter OD1. The second portion **214** extends proximally from the first portion **212**. The second portion **214** is a cylindrical tube defining a portion of the central passageway **208**. The wall of the second portion **214** has a second wall thickness thicker than the first wall thickness such that an outer surface of the wall of the second portion defines a second outer diameter OD2 larger than the first outer diameter OD1. The change in wall thickness from the first portion **212** to the second portion **214** creates a shoulder **218** between the first portion **212** and the second portion **214**. The second portion **214** further includes threads **220** along a portion thereof.

[0111] The ratchet lock **216** is disposed proximal of the second portion **214**. The ratchet lock **216** includes a plurality of teeth **222** extending circumferentially. In particular, as best shown in FIG. **20** each tooth **222** includes a ramp **224** and a shoulder **226**. Each ramp **224** extends radially outwardly and circumferentially from a first end spaced a first radius R1 from a central longitudinal axis CLA of the passageway **208** to a second end spaced a second radius R2 from the central longitudinal axis CLA of the passageway **208**. The shoulder **226** extends from the second end substantially radially inward toward the central longitudinal axis CLA. The ratchet lock **216** may further include a shoulder **228** extending radially outwardly from the second portion **214**, as shown in FIG. **20**.

[0112] The cap **250** of the introducer hub **200** is configured to couple the sheath **110** to the hub body **202**, as described in more detail below. In the embodiment shown, the cap **250** includes a cap body **252** including a proximal end **254**, a distal end **256**, and a central lumen **258** extending from the proximal end **254** to the distal end **256**. The cap **250** is a substantially cylindrical, stepped tube. Thus, in the embodiment shown, the cap body **252** includes a first longitudinal portion **260**, a second longitudinal portion **262** proximal of the first longitudinal portion **260**, a third longitudinal portion **264** proximal of the second longitudinal portion **262**, and a fourth longitudinal portion **266** proximal of the third longitudinal portion **264**. The first, second, third, and fourth longitudinal portions **260**, **262**, **264**, **266** are defined by having different internal diameters. Thus, the first longitudinal portion **260** has an inner surface defining a portion of the central lumen **258** and having a first inner diameter ID1, the second longitudinal portion **262** has an inner surface defining a portion of the central lumen **258** and having a second inner diameter ID2, the third longitudinal

portion **264** has an inner surface defining a portion of the central lumen **258** and having a third inner diameter ID3, and the fourth longitudinal portion **266** has an inner surface defining a portion of the central lumen **258** and having a fourth inner diameter ID4. In the embodiment shown, the first inner diameter ID1 is smaller than the second inner diameter ID2, which is smaller than the third inner diameter ID3, which is smaller than the fourth inner diameter ID4. Further, a first shoulder **261** is disposed at the inner surface between the first longitudinal portion **260** and the second longitudinal portion **262**, a second shoulder **263** is disposed at the inner surface between the second longitudinal portion **262** and the third longitudinal portion **264**, and a third shoulder **265** is disposed at the inner surface between the third longitudinal portion **264** and the fourth longitudinal portion **266**.

[0113] As explained in more detail below, the cap **250** is configured to capture a proximal end of the sheath **110** between the cap body **252** and the hub body **202**. Thus, in the embodiment shown, the first inner diameter ID1 of the cap body **252** is smaller than the first outer diameter OD1 of the hub body **202**, the second inner diameter ID2 of the cap body **252** is larger than the first outer diameter OD1 of the hub body **202** but smaller than the second outer diameter OD2 of the hub body **202**, the third inner diameter ID3 of the cap body **252** is larger than the second outer diameter of the hub body **202** but smaller than an outer diameter of the hub body **202** at the ratchet lock **216**, and the fourth inner diameter ID4 of the cap body is larger than the outer diameter of the hub body **202** at the ratchet lock **216**, as shown in FIG. **21**.

[0114] Further, the inner surface of the cap body **252** at the third longitudinal portion **264** includes inner threads **270** configured to mate with the threads **220** on the outer surface of the second portion **214** of the hub body **202**. The inner surface of the cap body **252** at the fourth longitudinal portion **266** includes two teeth **272** (see FIG. **18**) configured to lock onto the teeth **222** of the ratchet lock **216** of the hub body **202**. Although the particular embodiment shows two teeth **272**, this is not meant to be limiting, and more or fewer teeth **272** may be utilized in keeping with the objects of the present description. As shown in FIG. **18**, the outer surface of the cap body **252** may include ridges **274** or other similar features to assist a user in gripping the cap **250** to advance the cap **250** proximally over the proximal portion of the sheath **110** and the distal portion of the hub body **202**, as described below.

[0115] With reference to FIGS. **22-25**, a method of coupling the sheath **110** to the introducer hub **200** will be described. In a step of the method, a flared proximal portion **111** of the sheath **110** is placed over the first portion **212** of the hub body **202**, as shown in FIG. **22**.

[0116] In another step of the method, as shown in FIG. **23**, the cap **250** is placed over the sheath **110**. In particular, the proximal end **254** of the cap **250** may be placed over the distal end of the sheath **110** and the cap **250** may be advanced proximally towards the hub body **202**. In another step of the method, as shown in FIG. **24**, the cap **250** is rotated relative to the hub body **202** such that the threads **270** of the cap **250** advance along and engage the threads **220** of the hub body **202**. The cap **250** continues to be rotated and advanced proximally until the teeth **272** of the cap **250** lock onto two of the teeth **222** of the ratchet lock **216**, as shown in FIG. **25**. The cap **250** is thereby locked onto the hub body **202** and couples the sheath **110** to the hub **200**, as shown in FIGS. **19** and **21**.

[0117] FIGS. **26-29** show devices and methods for securing a sheath of an introducer to an introducer hub **300** of the introducer. In embodiments, the sheath may be the sheath **110** as described above or may be other sheaths. In embodiments, the introducer hub **300** may be similar to the introducer hub **150** described above except for details described herein with respect to the introducer hub **300**. However, this is not meant to be limiting, and the introducer hub **300** is not limited to the details described above with respect to the introducer hub **150**.

[0118] FIG. **26** shows a longitudinal cross-sectional view of the introducer hub **300** with the sheath **110** coupled thereto according to embodiments herein. Those skilled in the art will recognize that all the details of the introducer hub **300** are not described and that other features may be included

with the introducer hub **300** or that features may be excluded or modified. The introducer hub **300** includes a body **302** having a proximal end **304**, a distal end **306**, and a central lumen or passageway **308** extending from the proximal end **304** to the distal end **306**. The introducer hub **300** may include seals, seal caps, and other features as described above or as known to those skilled in the art. The introducer hub **300** further includes a cap **350** for coupling the sheath **110** to the hub body **302**.

[0119] In the embodiment shown, the hub body **302** of the introducer hub **300** includes a distal tube **310** extending proximally from the distal end **306** of the hub body **302**. The distal tube **310** includes a first portion **312**, a second portion **314** proximal of the first portion **312**, and a third portion **316** proximal of the second portion **314**. In particular, as shown in FIG. 26, the first portion **312** is a cylindrical tube with an inner surface thereof defining a portion of the central passageway **308**. The wall of the first portion has a first wall thickness such that an outer surface thereof defines a first outer diameter OD1. The second portion **314** extends proximally from the first portion **312**. The second portion **312** is a cylindrical tube defining a portion of the central passageway **308**. The wall of the second portion **314** has a second wall thickness thicker than the first wall thickness such that an outer surface of the wall of the second portion defines a second outer diameter OD2 larger than the first outer diameter OD1. The change in wall thickness from the first portion **312** to the second portion **314** creates a shoulder **318** between the first portion **312** and the second portion **314**. The second portion **314** further includes threads **320** along a portion thereof.

[0120] The third portion **316** is disposed proximal of the second portion **314**. The third portion **314** is a cylindrical tube defining a portion of the central passageway **308**. The wall of the third portion **314** has a second wall thickness thicker than the second wall thickness such that an outer surface of the wall of the third portion defines a third outer diameter OD3 larger than the second outer diameter OD2. The change in wall thickness from the second portion **314** to the third portion **316** creates a shoulder **317** between the second portion **314** and the third portion **316**. At a proximal end of the third portion **316**, the wall thickness is reduced, thereby creating a lip **319** for the cap **350** to lock onto, as described below.

[0121] The cap **350** of the introducer hub **300** is configured to couple the sheath **110** to the hub body **302**, as described in more detail below. In the embodiment shown, the cap **350** includes a cap body **352** including a proximal end **354**, a distal end **356**, and a central lumen **358** extending from the proximal end **354** to the distal end **356**. The cap **350** is a substantially cylindrical, stepped tube. Thus, in the embodiment shown, the cap body **352** includes a first longitudinal portion **360**, a second longitudinal portion **362** proximal of the first longitudinal portion **360**, a third longitudinal portion **364** proximal of the second longitudinal portion **362**, and a fourth longitudinal portion **366** proximal of the third longitudinal portion **364**. The first, second, third, and fourth longitudinal portions **360**, **362**, **364**, **366** are defined by having different internal diameters. Thus, the first longitudinal portion **360** has an inner surface defining a portion of the central lumen **358** and having a first inner diameter ID1, the second longitudinal portion **362** has an inner surface defining a portion of the central lumen **358** and having a second inner diameter ID2, the third longitudinal portion **364** has an inner surface defining a portion of the central lumen **358** and having a third inner diameter ID3, and the fourth longitudinal portion **366** has an inner surface defining a portion of the central lumen **358** and having a fourth inner diameter ID4. In the embodiment shown, the first inner diameter ID1 is smaller than the second inner diameter ID2, which is smaller than the third inner diameter ID3, which is smaller than the fourth inner diameter ID4. Further, a first shoulder **361** is disposed at the inner surface between the first longitudinal portion **360** and the second longitudinal portion **362** and a second shoulder **363** is disposed at the inner surface between the second longitudinal portion **362** and the third longitudinal portion **364**. Further, a lip **368** extends radially inwardly from the fourth longitudinal portion **366** at the proximal end **354** of the cap **350**.

[0122] As shown in FIGS. 26 and 29, the fourth longitudinal portion **366** comprises a plurality of

legs **367** extending longitudinally from third longitudinal portion **364**. The legs **367** are separated circumferentially by gaps **369**. Thus, the legs **367** are flexible such that the legs **367** may flex radially inwardly and radially outwardly. Further, the inner surface of the third longitudinal portion **364** includes threads **370** configured to engage the threads **320** of the distal tube **310** of the hub body **302**.

[0123] As explained in more detail below, the cap **350** is configured to capture a proximal end of the sheath **110** between the cap body **352** and the hub body **302**. Thus, in the embodiment shown, the first inner diameter ID1 of the cap body **352** is smaller than the first outer diameter OD1 of the hub body **302**, the second inner diameter ID2 of the cap body **352** is larger than the first outer diameter OD1 of the hub body **302** but smaller than the second outer diameter OD2 of the hub body **302**, the third inner diameter ID3 of the cap body **352** is larger than the second outer diameter of the hub body **302** but smaller than the third outer diameter OD3 of the hub body **302**, and the fourth inner diameter ID4 of the cap body **352** is larger than the third outer diameter OD3 of the hub body **302**, as shown in FIG. 26.

[0124] With reference to FIGS. 28-29, a method of coupling the sheath **110** to the introducer hub **300** will be described. In a step of the method, the flared proximal portion **111** of the sheath **110** is placed over the first portion **314** of the hub body **302**, as shown in FIG. 28. In another step of the method, the cap **350** is placed over the sheath **110**. In particular, the proximal end **354** of the cap **350** may be placed over the distal end of the sheath **110** and the cap **350** may be advanced proximally towards the hub body **302**. In another step of the method, the cap **350** is rotated relative to the hub body **302** such that the threads **370** of the cap **350** advance along and engage the threads **320** of the hub body **302**. The cap **350** continues to be rotated and advanced proximally until the lips **368** at the proximal ends **354** of the legs **367** of the cap **350** extends proximally past the proximal end of the third portion **316** of the distal tube **310** and engage lip **319** of the distal tube **310**, as shown in FIGS. 26, 27, and 29. As the cap **350** is advanced proximally the legs **367** flex radially outwardly such that the lips **368** at the proximal end **354** of the legs **367** extend over the third portion **316** of the distal tube **310**. As the cap **350** continues to be advanced proximally, the lips **368** clear the proximal end of the third portion **316** of the distal tube **310**, and the legs **367** flex back radially inwardly to their original position such that the lips **368** of the legs **367** engage the lip **319** at the proximal end of the third portion **316** of the distal tube **310**. This engagement of the lips **368** with the lip **319** locks the cap **350** in place, preventing the cap **350** from moving distally, and captures the flared proximal portion **111** of the sheath **110** between the cap **350** and the distal tube **310** of the hub body **302**.

[0125] FIGS. 30-35 show devices and methods for securing a sheath of an introducer to an introducer hub **400** of the introducer. In embodiments, the sheath may be the sheath **110** as described above or may be other sheaths. In embodiments, the introducer hub **400** may be similar to the introducer hub **150** described above except for details described herein with respect to the introducer hub **400**. However, this is not meant to be limiting, and the introducer hub **400** is not limited to the details described above with respect to the introducer hub **150**.

[0126] FIG. 34 shows the introducer hub **400** with an introducer sheath **110** coupled thereto via a connector **450**. The connector **450** is shown in FIGS. 31 and 32 and includes a body or housing **452** having a proximal end **454**, a distal end **456**, and a central lumen **458** extending from the proximal end **454** to the distal end **456**. The housing **452** further includes a plurality of legs **470** adjacent the proximal end **454** formed via gaps **472** between the legs **470**. The gaps **472** enable the legs **470** to flex radially outwardly and inwardly, as discussed in more detail below. As best shown in FIG. 35, the proximal end **454** of the housing **452** includes a proximal lip **462** extending radially inward. Similarly, the distal end **456** of the housing **452** includes a distal lip **460** extending radially inwardly.

[0127] The connector **450** further includes a grab ring **440** as shown in FIG. 31. The grab ring **440** is a ring with a series of struts **442** extending radially outwardly and inwardly coupling first bends

disposed at radially inward ends of the struts **442** and second bends **446** disposed at radially outward ends of the struts **442**. The grab ring **440** is disposed in the lumen **458** of the body **452** adjacent to the distal end **456** when the connector **450** couples the sheath **110** to the introducer hub **400**.

[0128] In the embodiments, the introducer hub **400** includes a hub body **402** having a proximal end (not shown), a distal end **406**, and a central lumen **408** extending therebetween. The hub body **402** includes a distal tube **410** extending proximally from the distal end **406** of the hub body. The distal tube **410** is a generally cylindrical tube defining a portion of the central lumen **408**. The distal tube **410** further includes a first groove **412** extending circumferentially around an outer surface of the distal tube **410**. The groove **412** extends radially inward from the outer surface of the distal tube **410**. The distal tube **410** further includes a second groove **414** extending circumferentially around an inner surface of the distal tube **410**. The second groove **414** extends radially outward from the inner surface of the distal tube **410** and is spaced distally of the first groove **412**, as shown in FIG. 35.

[0129] The introducer of FIGS. 30-35 further includes an O-ring **430** as shown in FIG. 31.

[0130] With the parts of the introducer described, the connection between the sheath **110** and the introducer hub **400** will be described with reference to FIGS. 34 and 35. The O-ring **430** is disposed in the second groove **414** of the distal tube **410** and the proximal end of the sheath **110** is disposed within the distal tube **410** and within the O-ring **430**. The O-ring **430** seals against the outer surface of the sheath **110** to prevent leakage between the sheath **110** and the inner surface of the distal tube **410**.

[0131] The connector **450** with the grab ring **440** disposed in a distal portion of the central lumen **458** as advanced proximally over the sheath **110**. As the housing **452** continues to be advanced proximally, the housing **452** extends over the distal tube **410** of the introducer hub **400**. Because the proximal lip **462** extends radially inwardly, the legs **470** flex radially outwardly such that the housing **452** can be advanced over the distal tube **410**. The housing **452** is advanced proximally until the proximal lip **462** reaches the first groove **412** of the distal tube **410**, at which point the legs **470** flex inwardly back to their original position, thereby locking the housing **452** longitudinally with respect to the distal tube **410** via the proximal lip **462** extending into the first groove **412**. The distal lip **460** of the housing **452** extends radially inwardly to contact the sheath **110**. The distal lip **460** also presses the grab ring **440** against the distal end **406** of the distal tube **410**.

[0132] FIGS. 36-38 show devices and methods for securing a sheath of an introducer to an introducer hub **500** of the introducer. In embodiments, the sheath may be the sheath **110** as described above or may be other sheaths. In embodiments, the introducer hub **500** may be similar to the introducer hub **150** described above except for details described herein with respect to the introducer hub **500**. However, this is not meant to be limiting, and the introducer hub **500** is not limited to the details described above with respect to the introducer hub **150**.

[0133] FIG. 36 shows an exploded view of the introducer hub **500** and the sheath **110**. Those skilled in the art will recognize that all the details of the introducer hub **500** are not described and that other features may be included with the introducer hub **500** or that features may be excluded or modified. Generally, the introducer hub **500** includes a hub body **502** and a locking clip **550**. The hub body **502** defines a proximal end **504**, a distal end **506**, and a central lumen or passageway **508** extending from the proximal end **504** to the distal end **506**. The introducer hub **500** may include seals, seal caps, and other features as described above or as known to those skilled in the art. The locking clip **550** is configured to couple the sheath **110** to the hub body **502**, as described below. In embodiments, the locking clip **550** may be formed of GRILAMID or a similar material.

[0134] In the embodiment shown, the hub body **502** of the introducer hub **500** includes a distal tube **510** extending proximally from the distal end **506** of the hub body **502**. The distal tube **510** may be a generally cylindrical tube with an inner surface defining a portion of the central passageway **508**. An inner diameter D1 of the distal tube **510** or lumen diameter D1 of the central passageway **508** at

the distal tube **510** is configured to receive the sheath **110** and the locking clip **550**, as described below. The distal tube **510** further includes at least one slot **512** extending from an outer surface through the inner surface of the distal tube **510**. The slot **512** is oriented circumferentially. In the embodiment shown, there are two slots **512** disposed opposite each other. However, this is not meant to be limiting, and in other embodiments there may be more or fewer slots **512**.

[0135] The locking clip **550** is configured to couple the sheath **110** to the hub body **502**, as described in more detail below. In the embodiment shown, the locking clip **550** is a generally cylindrical tube including a clip body **552** defining a proximal end **554**, a distal end **556**, and a central lumen **558** extending from the proximal end **554** to the distal end **556**. The locking clip **550** further includes a lip **570** extending radially outward at the distal end **556** of the clip body **552**. The locking clip **550** further includes at least one clip arm **560** proximal of the lip **570**. In the embodiment shown, the locking clip **550** includes two clip arms **560** disposed opposite each other to match the slots **512** of the distal tube **510**. A small portion of the second clip arm **560** can be seen in FIG. 36. The clip arms **560** are formed of a portion of the clip body **552** configured to flex radially inwardly and outwardly in reaction to forces acting thereupon. In particular, each clip arm **560** may be formed as a partially cut-out of the clip body **552**. In the embodiment shown, a first longitudinal cut **562** defines a first longitudinal edge of the clip arm **560**, a distal circumferential cut **564** defines a distal edge of the clip arm **560**, and a second longitudinal cut (not shown) opposite the first longitudinal cut **562** defines a second longitudinal edge of the clip arm **560**. A proximal end of the clip arm **560** is coupled to the remainder of the clip body **552**. Thus, the cantilever design of the clip arm **560** enables the clip arm **560** to flex radially inwardly and outwardly in response to forces acting thereupon. The clip arm **560** further includes an arm lip **566** extending radially outward at the distal end of the clip arm **560**. The arm lip **566** extends circumferentially along the distal edge of the clip arm **560** and is configured to interact with one of the slots **512** of the hub body **502**, as explained below.

[0136] An outer surface of the clip body **552** of the locking clip **550** defines an outer diameter **D2**. An inner surface of the clip body **552** defines an inner diameter **D3** or lumen diameter **D3** of the central lumen **558**. An outer surface of the lip **570** defines an outer diameter **D4**. Further, outer surfaces of the arm lips **566** of clip arms **560** disposed opposite each other define an outer diameter **D5** that may be equal to or slightly smaller than the outer diameter **D4**. Further the outer diameter **D5** is larger than the inner diameter **D1** of the central lumen **508** of the distal tube **510**. Stated another way, an outer surface of one of the arm lips **566** of one of the clip arms **560** defines a radius from the central longitudinal axis of the central lumen **558**, and that radius may be approximately half the outer diameter **D5** or slightly larger than half the inner diameter **D1** of the central lumen **508** of the distal tube **510**.

[0137] The sheath **110** includes a sheath lip (flare proximal portion) **111** at a proximal end thereof. An outer surface of the sheath **110** defines an outer sheath diameter **D6** and an outer surface of the sheath lip **111** defines an outer sheath lip diameter **D7** that is larger than the outer sheath diameter **D6**.

[0138] With reference to FIGS. 27-28, a method of coupling the sheath **110** to the introducer hub **500** will be described. In a step of the method, the locking clip **550** is placed over the sheath **110**. In particular, the proximal end **554** of the clip body **552** is aligned with a distal end of the sheath **110** and the locking clip **550** is slid over the sheath **110** such that the sheath **110** is located within the central lumen **558** of the locking clip **550**. Thus, the outer sheath diameter **D6** is smaller than the lumen diameter **D3** of the central lumen **558** of the locking clip **550**. As indicated by arrow **A1**, the locking clip **550** and/or the sheath **110** are translated relative to each other such that the locking clip **550** translates proximally relative to the sheath **110**, as shown in FIG. 27. The sheath lip **111** at the proximal end of the sheath **110** prevents the locking clip **550** from translating proximal of the sheath lip **111**. In other words, in the embodiment shown, the outer sheath lip diameter **D7** is larger than the lumen diameter **D3** of the central lumen **558** of the locking clip **550** such that the proximal

end **554** of the locking clip abuts against the sheath lip **111**.

[0139] In another step of the method, the sheath **110** and locking clip **550** are translated proximally relative to the hub body **502** as indicated by the arrows **A2** in FIG. **38** such that the proximal ends of the sheath **110** and the locking clip **550** are inserted into the central lumen **508** of the distal tube **510**. Those skilled in the art will recognize that the translation is directed to relative movement, so the sheath/locking clip **110/550** can be translated proximally, the hub body **501** can be translated distally, or some combination of proximal and distal translation may be utilized. With the proximal ends of the sheath **110** and the locking clip **550** being inserted into the central lumen **508** of the hub body **502**, the outer sheath lip diameter **D7** and the outer clip diameter **D2** are both smaller than the lumen diameter **D1** of the central lumen **508** at the distal tube **510**. However, the outer diameter **D5** at the arm lips **566** is larger than the lumen diameter **D1** of the central lumen **508** of the distal tube **510**. Thus, the clip arms **560** flex radially inwardly such that the sheath **110**/locking clip **550** combination may be inserted into the central lumen **508** of the distal tube **510**. When the sheath **110**/locking clip **550** combination is inserted into the central lumen **508** of the distal tube **510** a defined distance, the arms lips **566** of the clip arms **560** align with the slots **512** of the distal tube **510**, thereby enabling the clip arms **560** to flex back to the original radial position as the arm lips **566** extend radially into the slots **512**. The relative longitudinal locations of the clip arms **560**, arms lips **566**, and slots **512** are such that when the arm lips **566** extend into the slots **512**, the lip **570** at the distal end **556** of the locking clip **550** abuts against the distal end **506** of the distal tube **510**. Thus, the outer diameter **D4** of the lip **570** of the locking clip **550** is larger than the lumen diameter **D1** of the central lumen **508** of the distal tube **510**. With the method as described completed, the sheath **110** is coupled to the distal tube **510** of the hub body **502** and locked into place by the locking clip **550**.

[0140] FIGS. **39-41** devices and methods for securing a sheath of an introducer to an introducer hub **600** of the introducer. In embodiments, the sheath may be the sheath **110** as described above or may be other sheaths. In embodiments, the introducer hub **600** may be similar to the introducer hub **150** described above except for details described herein with respect to the introducer hub **600**. However, this is not meant to be limiting, and the introducer hub **600** is not limited to the details described above with respect to the introducer hub **150**.

[0141] FIG. **39** shows a perspective view of the introducer hub **600** with the sheath **110** coupled thereto by a connector **650**. FIG. **40** shows a longitudinal cross-sectional view of the sheath **110** coupled to the introducer hub **600** by the connector. FIG. **41** shows a perspective view of a longitudinal cross-section of the sheath **110** coupled to the introducer hub **600** via the connector **650**. Those skilled in the art will recognize that all the details of the introducer hub **600** are not described and that other features may be included with the introducer hub **600** or that features may be excluded or modified.

[0142] Generally, the introducer hub **600** includes a hub body **602** defining a proximal end **604**, a distal end **606**, and a central lumen or passageway **608** extending from the proximal end **604** to the distal end **606**. The introducer hub **600** may include seals, seal caps, and other features as described above or as known to those skilled in the art. The connector **650** is configured to couple the sheath **110** to the hub body **602**, as described below. Further, a band or strap **670** surrounds a portion of the connector **650** to couple the connector to the introducer hub **600**.

[0143] In the embodiment shown, the hub body **602** of the introducer hub **600** includes a distal tube **610** extending proximally from the distal end **606** of the hub body **602**. The distal tube **610** may be a generally cylindrical tube with an inner surface defining a portion of the central passageway **608**. In the embodiment of FIGS. **39-41**, the distal tube **610** includes barbs **612** extending radially outwardly along the distal tube **610**. In the embodiment shown, the distal tube includes three barbs **612** extending radially outwardly and circumferentially around the circumference of the distal tube **610**. In particular, each barb **612** includes a first, distal end **614**. A ramp **616** extends radially outwardly and proximally from the first, distal end **614** to a second, proximal end **618** of the barb

612. At the second, proximal end **614**, the barb **612** extends radially inwardly, forming a shoulder **620** at the proximal end of the barb **612**.

[0144] The connector **650** is configured to couple the sheath **110** to the hub body **602**, as described in more detail below. In the embodiment shown, the connector **650** is a generally cylindrical tube including a body **652** defining a proximal end **654**, a distal end **656**, and a central lumen **658** extending from the proximal end **654** to the distal end **656**. The connector **650** includes a proximal portion **660**, a distal portion **662**, and a lip **664** between the proximal portion **660** and the distal portion **662**. The central lumen **658** is continuous through the proximal portion **660** and the distal portion **662**.

[0145] As best shown in FIGS. **40** and **41**, the distal portion **662** of the connector **650** is configured to receive a proximal portion of the sheath **110**. In particular, the distal portion **662** of the connector **650** is overmolded over the proximal portion of the sheath **110**. Thus, in embodiments herein, the material of the connector **650**, or at least the distal portion **662** of the connector **650**, is the same as or compatible with the material of an outer layer of the proximal portion of the sheath **110** such that overmolding will attach the proximal portion of the sheath **110** to the distal portion **662** of the connector **650**. In a non-limiting example, the material of the connector **650** may be a polyether-based thermoplastic polyurethane such as, but not limited to, Texin® 85A. However, other suitable materials known to those skilled in the art may also be used.

[0146] As also shown in FIGS. **40** and **41**, the proximal portion **660** of the connector **650** is disposed over the distal tube **610** of the hub body **602**. In particular, the proximal portion **660** of the connector **650** extends proximally over the distal tube **610** from the distal end **606** of the distal tube **610** and covers the barbs **612** of the distal tube **610**. The proximal end **654** of the connector **650** may include a ridge **655** for increased strength of the connector at the proximal end **654**.

[0147] The band **670** is configured to be disposed around the proximal portion **660** of the connector **650** to couple the connector **650** to the distal tube **610** of the hub body **602**. In an embodiment, the band **670** is a generally cylindrical tube including a proximal end **672**, a distal end **674**, and a lumen **676** extending from the proximal end **672** to the distal end **674**. The lumen **676** is sized such that the band **670** applies a radially inward force on the proximal portion **660** of the connector **650** to deform the material of the proximal portion **660** onto the barbs **612** of the distal tube **610**. In a non-limiting example, the band **670** may be formed of stainless steel and the proximal portion **660** of the connector **650** may be formed of a polyether-based thermoplastic polyurethane such as, but not limited to, Texin® 85A. However, those skilled in the art will recognize that other materials may be used.

[0148] To attach the sheath **110** to the introducer hub **600**, the distal portion **662** of the connector **650** is overmolded over the proximal portion of the sheath **110**. The proximal portion **660** of the connector **650** is translated over the distal tube **610** of the hub body **602**. The orientation of the barbs **612** extending proximally assists in the proximal portion **660** extending over the barbs **612**. The band **670** is then translated proximally over the proximal portion **660** of the connector **650**, applying an inward force to the proximal portion **660**. The orientation of the barbs **612** as described above assists in the band **670** advancing proximally. Further, once connected, the orientation of the barbs **612** resists distal movement of the proximal portion **660** of the connector **650** such that the connection between the sheath **110** and the hub **600** is secure.

[0149] FIGS. **42A-50** show a dilator **700** according to embodiments hereof. The dilator **700** is configured to be used with an expandable introducer, such as, but not limited to the expandable introducers described herein, and in particular, to the expandable sheath described with respect to FIGS. **1-16** hereof. However, this is not meant to be limiting and the dilator **700** may be used with other expandable introducers and other expandable sheaths thereof. Those skilled in the art will recognize that features may be added and/or removed from the dilator **700**.

[0150] The dilator **700** generally includes a dilator shaft **702** coupled to a dilator hub **704**. Details of the dilator hub **704** and the connector of the dilator shaft **702** to the dilator hub **704** are not

described herein. Accordingly, dilator hubs known to those skilled in the art may be utilized with the dilator shaft **702** described herein.

[0151] The dilator shaft **702** includes a proximal end **706** for coupling to the dilator hub **704**, a distal end **708**, and a central lumen **710** extending from the proximal end **706** to the distal end **708**. The dilator shaft **702** further includes a proximal portion **712**, a proximal transition portion **714**, a middle portion **716**, and a distal portion **718**. The lengths and diameters of the portions described herein are configured to interact with an expandable sheath such as the expandable sheath **110** described above. Accordingly, lengths and/or diameters described herein may be described as relative to portions of the expandable sheath **110** such that different sized dilators may be used with different sized expandable sheaths.

[0152] The proximal portion **712** of the dilator shaft **702** is configured to have a maximum outer diameter while still fitting within the introducer hub such as the introducer hub **150** described above. In an embodiment, the outer diameter of the proximal portion **712** of the dilator shaft **702** may be about 6 mm for a nominal 18 French dilator. Further, in an embodiment, the proximal portion **712** may have a length of about 200 mm.

[0153] The middle portion **716** of the dilator shaft **702** has a reduced outer diameter as compared to the proximal portion **712**. Thus, in an embodiment, the middle portion **716** has an outer diameter of about 4 mm for an 18 French dilator. However, as noted above, the diameters may vary depending on the application of the dilator and the introducer used. Thus, in an embodiment, the outer diameter of the middle portion **716** of the dilator shaft **702** is smaller than an inner diameter **D3** of the introducer sheath **110** when the introducer sheath **110** is in the radially unexpanded, folded state described above. Thus, in the embodiment described, the middle portion **716** of the dilator shaft **702** will not radially expand the expandable sheath **110** when the middle portion **716** is disposed within the expandable sheath **110**. In an embodiment, the middle portion **716** of the dilator shaft **702** may have a length of about 285 mm.

[0154] The proximal transition portion **714** is disposed between the proximal portion **712** and the middle portion **716**. The outer diameter of proximal transition portion **714** tapers in a distal direction from the outer diameter of the proximal portion **712** at the proximal end of proximal transition portion **714** to the outer diameter of the middle portion **716** at the distal end of the proximal transition portion **714**. Thus, in an example embodiment, the outer diameter of the proximal transition portion **714** tapers from about 6 mm at its proximal end to about 4 mm at its distal end over a length of about 30 mm from its proximal end to its distal end. However, as explained above, this is not meant to be limiting.

[0155] The distal portion **718** of the dilator shaft **702** extends distally from the middle portion **716**. As shown in FIG. 42B, the distal portion **718** may comprise a distal transition portion **720**, a distal cylindrical portion **722**, and a distal tapered tip **724**. The distal transition portion **720** flares distally from the outer diameter of the middle portion **716** to the outer diameter of the distal cylindrical portion **722**. The distal cylindrical portion **722** extends distally from the distal transition portion **720**. The distal cylindrical portion **722** is sized such that when the distal cylindrical portion **722** is retracted proximally through the expandable sheath **110**, the distal cylindrical portion **722** pre-dilates or conditions the expandable sheath **110** for radially expansion, as shown in FIGS. 48-50. Thus, in an example, the outer diameter of the distal cylindrical portion **722** approximately equals the outer diameter of the expandable sheath **110** in the radially unexpanded, folded state. In an example, the outer diameter of the distal cylindrical portion **722** is about 5.8 mm. In an example, the distal cylindrical portion **722** has a length of about 30 mm.

[0156] The distal tapered tip **724** extends distally from the distal cylindrical portion **722**. The distal tapered tip **724** tapers in a distal direction from a first outer diameter at its proximal end to a second outer diameter smaller than the first outer diameter at its distal end. The distal tapered tip **724** is configured for atraumatic insertion of the dilator into a vessel. In an embodiment, the outer diameter of the distal tapered tip **724** at its proximal end is about 5.8 mm and the outer diameter of

the distal tapered tip at its distal end is about 1.28 mm. Further, in an embodiment, the distal tapered tip **724** has a length of about 55 mm.

[0157] The central lumen **710** is defined by an inner diameter of the shaft **702**. The central lumen **710** may have a relatively constant diameter through the proximal portion **712**, the proximal transition portion **714**, the middle portion **716**, and a portion of the distal portion **718**. In an embodiment, the diameter of the central lumen is about 1.5 mm. In an embodiment, a distal portion of the central lumen **710** tapers in a distal direction. In an example, a distal 50 mm of the central lumen **710** tapers from a diameter of about 1.5 mm to a diameter of about 0.94 mm. Then, a final 12 mm of the central lumen remains a constant diameter of about 0.94 mm. Those skilled in the art will recognize that the sizes noted are merely examples of a particular embodiment of the dilator shaft **702**, and are not meant to be limiting.

[0158] Having described the dilator shaft **702** with respect to the expandable sheath **110**, in an embodiment, the proximal portion **712** and the proximal transition portion **714** of the dilator shaft **702** may initially not be coupled to the middle portion **716**. Thus, in such an embodiment, the middle portion **716** may be inserted into a distal end of the expandable sheath **110** and advanced proximally therethrough. The proximal transition portion **714**, the proximal portion **712**, and the dilator hub **704** may then be coupled to the middle portion **716**. In such a method, the introducer sheath **110** may remain in the radially unexpanded, folded state during insertion into a patient's vessel, as shown in FIG. **45**, and will be radially expanded only upon proximal retraction of the dilator shaft **702** through the expandable sheath **110**.

[0159] FIGS. **46** and **47** show the dilator shaft **702** disposed through the introducer sheath **110** with the distal “bump” (proximal portion of the distal cylindrical portion **722**) disposed within the tip **130** of the expandable sheath **110**. By placing the distal bump within the tip **130**, the tip **130** stretches. The stretching of the tip **130** provides a smaller profile of the tip **130**, thereby reducing the likelihood of the tip **130** catching within the vessel. In other words, the distal end of the tip **130** is stretched such that the distal end presents a thinner edge than when the distal end of the tip **130** is not stretched.

[0160] FIGS. **51-58** show distal transition portions of a dilator, such as the dilator **700** described previously. In the embodiments herein, the distal transition portions are configured to smooth or flatten the dilator to sheath transition, thereby providing a reduced likelihood of a tip of the sheath catching on tissue within the vessel and improving atraumaticity of the dilator to introducer sheath transition.

[0161] FIGS. **51-52** show the distal transition portion **720** of the dilator **700**, according to an embodiment of the present disclosure. The distal transition portion **720** includes a proximal or a first section **730** and a distal or a second section **740**. The first section **730** of the distal transition portion **720** includes a proximal end **732** and a distal end **734**, as shown in FIG. **51**. The proximal end **732** is disposed adjacent to a distal end of the middle portion **716** of the dilator **700** and extends distally to the distal end **734**. In embodiments herein, a length of the first section **730** of the distal transition portion **720** may be in the range of 2 mm to 15 mm. The first section **730** of the distal transition portion **720** has a first outer diameter OD1 at the proximal end **732** and a second outer diameter OD2 at the distal end **734**, wherein the first outer diameter OD1 is smaller than the second outer diameter OD2. The first outer diameter OD1 may be in the range of about 2 mm to about 10 mm. The second outer diameter OD2 may be in the range of about 2.5 mm to about 11 mm. An outer surface **736** of the first section **730** flares outwardly and distally in a concave profile from the first outer diameter OD1 at the proximal end **732** to the second outer diameter OD2 at the distal end **734**.

[0162] The second section **740** of the distal transition portion **720** includes a proximal end **742** and a distal end **744**. The proximal end **742** is disposed adjacent to the distal end **734** of the first section **730**. The second section **740** extends distally from the proximal end **742** to the distal end **744**. The distal end **744** is disposed adjacent to a proximal end of the distal cylindrical portion **722** of the

distal portion **718** of the dilator **700**. The length of the second section **740** of the distal transition portion **720** may be in the range of 2 mm to 10 mm. The second section **740** has the second outer diameter OD2 at the proximal end **742** and a third outer diameter OD3 at the distal end **744**. In embodiments herein, the second outer diameter OD2 is smaller than the third outer diameter OD3. The third outer diameter OD3 may be in the range of about 3 mm to about 12 mm. An outer surface **746** of the second section **740** flares outwardly and distally in a convex profile from the second outer diameter OD2 at the proximal end **742** to the third outer diameter OD3 at the distal end **744**. [0163] Thus, as shown in FIG. **51**, the distal transition portion **720** flares radially outward and is curved. As described, the first section **730** curves such that the outer surface **736** is concave when viewed from outside of the dilator **700** and the second section **740** curves such that the outer surface **746** is convex when viewed from outside of the dilator **700**. The embodiment of FIGS. **51-52** may also be referred to as rounded.

[0164] FIG. **52** shows a distal portion of the introducer sheath **110** disposed on an outer surface of the dilator **700**, with the tip **130** of the introducer sheath **110** disposed on the first section **730** and a portion of the second section **740** of the distal transition portion **720** of the dilator **700**. By placing the tip **130** on the first section **730** and a portion of the second section **740** of the distal transition portion **720** of the dilator **700**, the transition between the tip **130** and the second section **740** of the distal transition portion **720** of the dilator **700** is generally smooth. Thus, the rounded shape of the distal transition portion **720** of the dilator **700** provides a reduced likelihood of the tip **130** of the introducer sheath **110** catching on tissue within the vessel.

[0165] FIGS. **53-54** show a distal transition portion **820** according to another embodiment hereof. The distal transition portion **820** may be used with embodiments of the dilator **700** described above. In the embodiment of FIGS. **53-54**, the distal transition portion **820** includes a proximal end **832** and a distal end **834**. The proximal end **832** is disposed adjacent to the distal end of the middle portion **716** of the dilator **700** and extends distally to the distal end **834**, which is disposed adjacent to a proximal end of the distal cylindrical portion **722** of the distal portion **718** of the dilator **700**. In embodiments herein, a length of the distal transition portion **820** may be in the range of 2 mm to 20 mm. The distal transition portion **820** has a first outer diameter OD1 at the proximal end **832** and a second outer diameter OD2 at the distal end **834**, wherein the first outer diameter OD1 is smaller than the second outer diameter OD2. In embodiments herein, the outer surface **836** of the first section **830** is flared from the proximal end **832** to the distal end **834**. The first outer diameter OD1 may be in the range of about 2 mm to about 10 mm. The second outer diameter OD2 may be in the range of 3 mm to 12 mm. In the embodiment of FIGS. **53-54**, the outer surface **836** of the distal transition portion **820** flares linearly from the proximal end **832** to the distal end **834**. The distal transition portion **820** may also be referred to as a straight or linear transition portion.

[0166] FIG. **54** shows a distal portion of the introducer sheath **110** disposed on an outer surface of the dilator **700**. The tip **130** of the introducer sheath **110** is disposed on the distal transition portion **820** of the dilator **700**. Placing the tip **130** on the distal transition portion **820** proximal of the distal cylindrical portion **722** provides a generally smooth transition between the tip **130** and the distal cylindrical portion **722** of the dilator **700** and thereby reduces the likelihood of the tip **130** catching tissue within the vessel.

[0167] FIGS. **55-56** show a distal transition portion **920** according to another embodiment hereof. The distal transition portion **920** may be used with embodiments of the dilator **700** described above. In the embodiment of FIGS. **55-56**, the distal transition portion **920** includes a first section **930**, a second section **940**, a third section **950**, a fourth section **960**, and a fifth section **970**.

[0168] The first section **930** of the distal transition portion **920** includes a proximal end **932** and a distal end **934**. The proximal end **932** is disposed adjacent to the distal end of the middle portion **716** of the dilator **700** and extends distally to the distal end **934**. The length of the first section **930** of the distal transition portion **920** may be in the range of about 1 mm to about 5 mm. The first section **930** of the distal transition portion **920** has a first outer diameter OD1 at the proximal end

932 and a second outer diameter **OD2** at the distal end **934**. In embodiments herein, the first outer diameter **OD1** is smaller than the second outer diameter **OD2**. The first outer diameter **OD1** may be in the range of about 2 mm to about 10 mm. The second outer diameter **OD2** may be in the range of about 2.25 mm to about 10.5 mm. In embodiments herein, an outer surface **936** of the first section **930** flares outwardly and distally in a concave profile from the first outer diameter **OD1** at the proximal end **932** to the second outer diameter **OD2** at the distal end **934**.

[0169] The second section **940** of the distal transition portion **920** includes a proximal end **942** and a distal end **944**. The proximal end **942** is disposed adjacent to the distal end **934** of the first section **930**. The second section **940** extends distally from the proximal end **942** to the distal end **944**. The length of the second section **940** may be in the range of about 1 mm to about 5 mm. The second section **940** has the second outer diameter **OD2** at the proximal end **942** and a third outer diameter **OD3** at the distal end **944**. In embodiments herein, the second outer diameter **OD2** is smaller than the third outer diameter **OD3**. The third outer diameter **OD3** may be in the range of about 2.5 mm to about 11 mm. In embodiments herein, an outer surface **946** of the second section **940** flares outwardly and distally in a convex profile from the second outer diameter **OD2** at the proximal end **942** to the third outer diameter **OD3** at the distal end **944**.

[0170] The third section **950** of the distal transition portion **920** includes a proximal end **952**, disposed adjacent to the distal end **944** of the second section **940**, and extends distally to a distal end **954**. The length of the third section **950** may be in the range of about 1 mm to about 5 mm. The third section **950** includes a generally cylindrical outer surface **956** with the third outer diameter **OD3** extending from the proximal end **952** to the distal end **954**.

[0171] The fourth section **960** of the distal transition portion **920** includes a proximal end **962** and a distal end **964**. The proximal end **962** is disposed adjacent to the distal end **954** of the third section **950** and extends distally to the distal end **964**. The length of the fourth section **960** may be in the range of about 1 mm to about 5 mm. The fourth section **960** has the third outer diameter **OD3** at the proximal end **962** and a fourth outer diameter **OD4** at the distal end **964**. The third outer diameter **OD3** is smaller than the fourth outer diameter **OD4**. The fourth outer diameter **OD4** may be in the range of about 2.75 mm to about 11 mm. In embodiments herein, an outer surface **966** of the fourth section **960** flares outwardly and distally in a concave profile from the third outer diameter **OD3** at the proximal end **962** to the fourth outer diameter **OD4** at the distal end **964**.

[0172] The fifth section **970** of the distal transition portion **920** includes a proximal end **972** and a distal end **974**. The proximal end **972** is disposed adjacent to the distal end **964** of the fourth section **960**. The fifth section **970** extends distally from the proximal end **972** to the distal end **974**. The distal end **974** is disposed adjacent to the proximal end of the distal cylindrical portion **722** of the distal portion **718** of the dilator **700**. The length of the fifth section **970** may be in the range of about 1 mm to about 5 mm. The fifth section **970** has the fourth outer diameter **OD4** at the proximal end **972** and a fifth outer diameter **OD5** at the distal end **974**, wherein the fourth outer diameter **OD4** is smaller than the fifth outer diameter **OD5**. The fifth outer diameter **OD5** may be in the range of about 3 mm to about 12 mm. In embodiments herein, an outer surface **976** of the fifth section **970** flares outwardly and distally in a convex profile from the fourth outer diameter **OD4** at the proximal end **972** to the fifth outer diameter **OD5** at the distal end **964**. The embodiment of FIGS. 55-56 may be described as a double taper (in the proximal direction) or a double flare (in the distal direction).

[0173] FIG. 56 shows a distal portion of the introducer sheath **110** disposed on an outer surface of the dilator **700**, with the tip **130** of the introducer sheath **110** disposed on the distal transition portion **920** on first section **930**, the second section **940**, the third section **950**, and a portion of the fourth section **960**. In embodiments herein, utilizing the distal transition portion **920**, the transition between the tip **130** and the distal cylindrical portion **722** of the dilator **700** is generally smooth or flat to provide a reduced likelihood of the tip **130** catching or snagging tissue within the vessel.

[0174] FIGS. 57-58 show a distal transition portion **1020** according to another embodiment hereof.

The distal transition portion **1020** may be used with embodiments of the dilator **700** described above. As shown in FIG. 57, the distal transition portion **1020** includes a first or proximal section **1030**, a second or middle section **1040**, and a third or distal section **1050**.

[0175] The first section **1030** includes a proximal end **1032** and a distal end **1034**. The proximal end **1032** is disposed adjacent to the distal end of the middle portion **716** of the dilator **700**. The first section **1030** extends distally from the proximal end **1032** to the distal end **1034**. The length of the first section **1030** may be in the range of about 2 mm to about 5 mm. The first section **1030** has a first outer diameter OD1 at the proximal end **1032** and a second outer diameter OD2 at the distal end **1034**. In embodiments herein, an outer surface **1036** of the first section **1030** is flared linearly in a direction towards the distal end **1034** such that the first outer diameter OD1 at the proximal end **1032** is smaller than the second outer diameter OD2 at the distal end **1034**. The first outer diameter OD1 may be in the range of about 2 mm to about 10 mm. The second outer diameter OD2 may be in the range of about 2.5 mm to about 11 mm.

[0176] The second or middle section **1040** of the distal transition portion **1020** includes a proximal end **1042** disposed adjacent to the distal end **1034** of the first section **1030**. The second section **1040** extends distally from the proximal end **1042** to a distal end **1044**. The length of the second section **1040** may be in the range of about 1 mm to about 5 mm. The second section **1040** includes a generally cylindrical outer surface **1046** with the second outer diameter OD2 extending from the proximal end **1042** to the distal end **1044**.

[0177] The third or distal section **1050** of the distal transition portion **1020** includes a proximal end **1052** and a distal end **1054**. The proximal end **1052** is disposed adjacent to the distal end **1034** of the second section **1040** and extends distally to the distal end **1054**. The distal end **1054** is disposed adjacent to the proximal end of the distal cylindrical portion **722** of the distal portion **718** of the dilator **700**. The length of the third section **1050** may be in the range of about 1 mm to about 5 mm. The third section **1050** includes an outer surface **1056**. The outer surface **1056** is flared linearly from the proximal end **1052** to the distal end **1054** such that the second outer diameter OD2 at the proximal end **1052** is smaller than a third outer diameter OD3 at the distal end **1054**. The third outer diameter OD3 may be in the range of about 3 mm to about 12 mm.

[0178] The distal transition portion **1020** is similar to the distal transition portion **920** in that it includes a double taper or double flare. However, the distal transition portion **1020** includes linear flares instead of the double curved flares of the distal transition portion **920**.

[0179] As shown in FIG. 58, a distal portion of the introducer sheath **110** is disposed on an outer surface of the dilator **700**. In greater detail, the tip **130** of the introducer sheath **110** is disposed on the distal transition portion **1020** along first section **1030** and the second section **1040**. As shown in FIG. 58, utilizing the distal transition portion **1020**, the transition between the tip **130** and the distal cylindrical portion **722** of the dilator **700** is generally smooth or flat to provide a reduced likelihood of the tip **130** catching or snagging tissue within the vessel.

[0180] In embodiments herein, the distal transition portions **720**, **820**, **920**, and **1020** may each be formed of materials such as, but not limited to PTFE, FEP, Pebax, HDPE, LDPE or any other suitable material. Further, embodiments herein may include a lubricious coating on an outer surface thereof and may contain radiopaque markers such that the dilator and sheath tip relationship may be monitored using standard imaging techniques.

[0181] As used herein, the terms “generally” and “substantially” mean approximately. When used to describe angles such as “substantially parallel” or “substantially perpendicular” the term “substantially” means within 10 degrees of the angle. When used to describe shapes such as “substantially” or “generally” cylindrical or “substantially” or “generally” tube-shaped or “generally” or “substantially” conical, the terms mean that the shape would appear cylindrical or tube-shaped or conical to a person of ordinary skill in the art viewing the shape with a naked eye. The term “about” as used herein to refer to dimensions means within 5% of the dimension.

[0182] It should be understood that various embodiments disclosed herein may be combined in

different combinations than the combinations specifically presented in the description and accompanying drawings. It should also be understood that, depending on the example, certain acts or events of any of the processes or methods described herein may be performed in a different sequence, may be added, merged, or left out altogether (e.g., all described acts or events may not be necessary to carry out the techniques). In addition, while certain aspects of this disclosure are described as being performed by a single device or component for purposes of clarity, it should be understood that the techniques of this disclosure may be performed by a combination of devices or components.

Claims

1. An introducer comprising: an introducer hub, the introducer hub including a proximal end, a distal end, and a central lumen extending from the proximal end to the distal end, the introducer hub including a distal tube at a distal portion of the introducer hub, the distal tube defining a portion of the central lumen; a sheath coupled to the introducer hub and extending distally therefrom; and a connector coupling the sheath to the distal tube of the introducer hub.
2. The introducer of claim 1, wherein: the distal tube comprises a cylinder having a stepped outer surface and including tube threads on a portion of the stepped outer surface; the connector comprises a cap comprising a cylinder having a stepped inner surface and including cap threads on a portion of the stepped inner surface; and the cap is configured to capture a proximal end of the sheath between the stepped inner surface of the cap and the stepped outer surface of the distal tube as the cap threads engage with the tube threads.
3. The introducer of claim 2, wherein: the distal tube further comprises a ratchet lock proximal of the tube threads, the ratchet lock comprising a plurality of teeth extending radially outwardly and circumferentially around the distal tube; and the cap further comprises at least one tooth extending radially inwardly proximal of the cap threads, wherein the at least one tooth is configured to engage the plurality of teeth of the ratchet lock to lock the cap to the distal tube.
4. The introducer of claim 2, wherein: the distal tube comprises a tube lip extending radially outwardly proximal of the tube threads; the cap includes a plurality of longitudinal legs extending proximally such that a proximal end of the legs is proximal of the cap threads, the plurality of longitudinal legs disposed adjacently around a circumference of the cap and including gaps between circumferentially adjacent longitudinal legs, the plurality of longitudinal legs comprising a cap lips extending radially inwards from the proximal end of the longitudinal legs; and the plurality of longitudinal legs are configured to flex radially outwardly for the cap lips to extend over the tube lip and are configured to flex back radially inwardly when the leg lips extend proximal of the tube lip such that the leg lips lock the cap to the distal tube.
5. The introducer of claim 1, wherein the connector comprises a housing, a grab ring, and an O-ring, wherein the housing comprises a tube including distal lip extending radially inwardly from a distal end of the housing, a plurality of longitudinal legs at a proximal end of tube, the longitudinal legs separated circumferentially by a plurality of gaps between adjacent legs of plurality of longitudinal legs, and a proximal lip extending radially inward from a proximal end of the longitudinal legs; the distal tube includes a first groove extending radially inwardly from an outer surface of the distal tube and circumferentially around the distal tube and a second groove extending radially outwardly from an inner surface of the distal tube and circumferentially around the inner surface of the distal tube; and with a proximal end of the sheath disposed within the distal tube; the O-ring is disposed in the second groove of the distal tube between an inner surface of the distal tube and an outer surface of the sheath; and the housing is configured to be translated over the sheath and over the distal tube proximally such that: the longitudinal legs flex radially outwardly as the proximal lip extend over the distal tube until the proximal lip reaches the first groove such that the longitudinal legs flex back radially inwardly as the proximal lip is disposed

within the first groove; and the distal lip of the housing extends radially inwardly distal of the distal end of the distal tube and abuts the outer surface of the sheath.

6. The introducer of claim 1, wherein: the connector comprises a locking clip configured to be disposed over the sheath, wherein the locking clip comprises a generally cylindrical tube including at least one clip arm proximal of a distal end of the locking clip, wherein the at least one clip arm is configured to flex radially inwardly and outwardly and includes a clip lip extending radially outwardly from the clip arm; the distal tube includes at least one slot extending from an outer surface to an inner surface thereof; the sheath includes a proximal lip extending radially outwardly therefrom; and with the locking clip disposed over the sheath such that a proximal end of the locking clip is adjacent the proximal lip of the sheath, the sheath and locking clip are configured to be inserted into the distal tube such that the at least one clip arm flexes radially inwardly until the clip lip aligns with the slot of the distal tube such the at least one clip arm flexes back radially outwardly and the clip lip extends into the slot to couple the sheath and the locking clip to the distal tube.

7. The introducer of claim 1, wherein: the connector comprises a generally cylindrical tube including a distal portion overmolded over a proximal end of the sheath, a proximal portion configured to be disposed over the distal tube of the introducer hub, and band configured to be disposed around the proximal portion to couple the proximal portion to the distal tube of the introducer hub; the distal tube of the introducer hub includes barbs extending radially outwardly along the distal tube, wherein the band of the connector is configured to deform the proximal portion of the generally cylindrical tube of the connector to couple the generally cylindrical tube of the connector to the distal tube of the introducer hub; and each barb extends circumferentially around the distal tube, and wherein each barb includes a barb distal end, a ramp extending radially outwardly and proximally from the barb distal end to a barb proximal end, and a shoulder at the barb proximal end extending radially inwardly from the barb proximal end.

8. A system comprising: a dilator comprising a dilator hub and a dilator shaft coupled to and extending distally from the dilator hub, the dilator shaft including: a proximal portion having a first outer diameter; a middle portion having a second outer diameter; a proximal transition portion disposed between the proximal portion and the middle portion, the proximal transition portion tapering in a distal direction from the first outer diameter to the second outer diameter; and a distal portion extending distally from the middle portion, the distal portion including a distal cylindrical portion having a third outer diameter, wherein the first outer diameter is larger than the second outer diameter and the third outer diameter, and wherein the third outer diameter is larger than the second outer diameter.

9. The system of claim 8, wherein the dilator shaft further includes a distal transition portion disposed between the middle portion and the distal cylindrical portion, wherein the distal transition portion flares outwardly distally from the second outer diameter of the middle portion to the third outer diameter of the distal cylindrical portion.

10. The system of claim 9, wherein the distal transition portion flares linearly outwardly from a proximal end adjacent a distal end of the middle portion to a distal end of the distal transition portion adjacent a proximal end of the distal cylindrical portion.

11. The system of claim 9, wherein the distal transition portion includes a proximal section and a distal section, the proximal section flaring radially outwardly with a curved concave profile, and a distal section flaring outwardly with a curved convex profile.

12. The system of claim 9, wherein the distal transition portion includes a proximal section extending distally from the middle portion, a middle section extending distally from the proximal section, and a distal section distal extending distally from the middle section, wherein the proximal section flares outwardly in a distal direction, the middle section is generally cylindrical, and the distal section flares outwardly in a distal direction.

13. The system of claim 12, wherein the proximal section and the distal section flare linearly

outwardly.

14. The system of claim 12, wherein each of the proximal section and the distal section includes curved concave profile and a curved convex profile.

15. The system of claim 8, wherein the distal portion of the dilator shaft further includes a distal tapered tip extending distally from the distal cylindrical portion wherein the distal tapered tip includes an outer diameter than tapers distally.

16. The system of claim 8, wherein the system further comprises an introducer comprising an introducer hub and an expandable sheath coupled to and extending distally from the introducer hub, wherein the expandable sheath is expandable from a radially compressed state wherein the expandable sheath has a first inner sheath diameter and a first outer sheath diameter to a radially expanded state wherein the sheath has a second inner sheath diameter larger than the first inner sheath diameter and a second outer sheath diameter larger than the first outer sheath diameter.

17. The system of claim 16, wherein: the first outer diameter of the dilator shaft proximal portion is sized to fit within the introducer hub; the second outer diameter of the dilator shaft middle portion is smaller than the first inner sheath diameter of the sheath; and the third outer diameter of the dilator shaft distal cylindrical portion is approximately equal to the first outer sheath diameter of the sheath.

18. An expandable introducer sheath comprising: a shaft configured to be expanded from a radially unexpanded, folded state to a radially expanded, unfolded state in reaction to a device being extended through a central lumen of the expandable sheath, wherein the shaft includes: an inner liner; and an outer wall disposed radially outwardly of the inner liner, the outer wall having a variable thickness around a circumference of the shaft; and a tip coupled to a distal portion of the shaft, the tip including a first circumferential portion and a second circumferential portion, wherein the second circumferential portion is more flexible than the first circumferential such that the second circumferential portion is configured to radially expand when the in reaction to the device extending through tip.

19. The expandable introducer sheath of claim 18, wherein in the radially expanded, unfolded state, the shaft includes a thick-walled region and a thin-walled region, wherein the thin-walled region is configured to fold when the shaft is in the radially unexpanded, folded state.

20. The expandable introducer sheath of claim 18, wherein in the radially unexpanded, folded state, the shaft includes a folded region and an unfolded region, wherein the second circumferential portion is circumferentially aligned with the folded region.
